PROJECT DOCUMENTATION (MANDATORY)

1. Repository & Hosting

- All project material — code, documentation, media, and design files — will be pushed to the official RoboticsIITJ GitHub repository.
- All iteration phase items will also be included here.
- The repository will be organized into two top-level divisions plus a shared resource-guide folder, as below.

```
/ROOT
├── learning/iteration-phase/
├── final-product/
└── resource-guide/
```

2. Division I — Learning / Iteration Phase

Captures the full exploratory journey of the project, preserved for anyone who wants to understand how the team arrived at the final approach — not just what the final approach is.

2.1 Individual Iterations

- Sub-folder per team member (or per sub-team), containing their own attempts, experiments, and dead ends.
- Each member's folder should be self-contained enough that a reader can see what was tried without needing external context.

2.2 Approach Documentation

A structured document (or set of documents) covering every approach that was tried, narrowing down to the one finally adopted. Split into two parts:

Software Document

- All algorithmic / software approaches attempted (e.g. trajectory planning strategies, IK methods, path-smoothing techniques)
- Pros and cons of each
- Justification for the final approach selected

Hardware Document

- All electrical / electronic components discussed and tried (motors, drivers, sensors, controllers, etc.)
- For each component: what problem it was meant to solve, what issues were encountered, and its pros
- Justification for the final component selection

2.3 Testing Videos

- Videos of all tests conducted during the iteration phase.

- Each video accompanied by a short written explanation of what was being tested, the setup, and the outcome / observation.

3. Division II — Final Product

The polished, deployable outcome of the project.

3.1 Codebase

- Clean, final code base as actually deployed to the board.
- No experimental / dead code — only what is in production use.

3.2 Architecture Diagrams

- High-level system architecture (overall data / control flow, major subsystems).
- Progressively detailed diagrams down to low-level (module-level, circuit-level, or function-level as applicable).

3.3 README

- Clear instructions for using the final product.
- Written so a new reader can understand the system and continue development on it without needing to ask the original team.
- Should cover: setup, dependencies, how to run / deploy, how the code is structured, and known limitations or next steps.

3.4 Demo Videos

- Videos demonstrating different aspects / capabilities of the final product — separate videos highlighting distinct features or use cases, not just one overview video.

3.5 CAD Models (if applicable)

- All CAD files for the mechanical design of the system.

3.6 Anything Else Relevant

- Any additional material the team identifies as important to fully represent the project (e.g. BOM, wiring diagrams, calibration notes, known issues log) — included at the team's discretion.

4. Resource Guide (Shared Folder)

- A separate folder, outside both divisions, containing curated learning resources.
- Intent: give anyone approaching this class of project (planar arm control, trajectory planning, embedded deployment, etc.) a starting point for building the necessary concepts before diving into this repo's code.
- Should point to foundational material rather than just project-specific notes — the goal is to teach the prerequisites, not just document this one implementation.

THIS DOCUMENTATION IS REQUIRED SO THAT FUTURE MEMBERS CAN LEARN FROM THIS AND START BUILDING ON IT , NOT JUST EVERY TIME REATTEMPTING THE SAME THINGS